

stimulation (FES), magnetic seizure therapy (MST), vagus nerve stimulation (VNS) and deep transcranial magnetic stimulation (deep TMS).

tDCS is getting loads of attention from early adopters who rave about its potential, and scientists who are trying to unravel what it can and cannot do. It is a non-invasive form of brain stimulation that involves passing a current between electrodes on the scalp. The technique is one of a number of technologies being tested to see if it can enhance cognitive functions such as boosting the performance of drone pilots or image analysts.

According to the experts, the current is set to about 2mA, which is about 1000 times less than the electrical current that flows through a typical iPad charger, and only about 1/50th of that current makes it through the skull to the brain. The stimulation is said to last for about ten minutes, and is aimed at the right inferior frontal cortex and the right anterior temporal lobe, which are brain areas thought to be important for learning.

A mild stinging sensation is felt where the electrode is attached to the head. This is thought to be a normal feeling, but if the sensation continues, it is better to turn off the connection and try to get a better connection.

Next, people notice a slight taste of metal in their mouth, a common side effect of tDCS, according to experts.

Although this is an experimental method, it is widely regarded as safe and does not appear to cause any short-term harm. The most common side effects of stimulation include tingling or skin irritation and seeing phosphenes (light rings or spots not produced by actual light). Other side effects include fatigue, headache and itching under the electrode. In rare cases, people experience nausea or insomnia.

Some studies have found impressive gains in performance from tDCS,

while others suggest the technique has little effect. That has not stopped adventurous amateurs from testing home-made devices on themselves, though this is not recommended. Scientists are still investigating how tDCS works, but the reigning hypothesis is that it changes the excitability of a particular brain region. Depending on the direction of the current, stimulation could make neurons in a particular area of the brain more or less likely to fire.

Recent studies have focused on using tDCS to boost the performance of image analysts for the military. The study found that participants who received training and stimula-



Electrical stimulation devices at Oxford University can have currents placed on various parts of the brain (Image courtesy: www.bbc.com)

tion were about 25 per cent more accurate in identifying the objects than those without stimulation. Some studies claim to have found benefits in everything from mental arithmetic to memory; other studies have found no evidence of cognitive effects from a single session of tDCS.

Research on tDCS is not limited to seeking cognitive boosts. Many scientists are also investigating its efficacy in treating mental disorders such as traumatic brain injury and schizophrenia. However, there is no administrative approval for the use of the technique.

Similarly, TMS uses a magnetic coil to induce small electrical currents in the brain. In contrast to tDCS, TMS actually causes neurons to send signals or spike, and it has

been approved for treating depression and migraines.

Effects of electric brain stimulation

Focused thinking, better memory, deeper sleep, relief from depression and reduced stress are some benefits of electronic stimulation for the mind and brain as mentioned in the literature on the Internet by companies dealing with the development of different types of the brain stimulation devices.

Popular electric brain stimulation methods that are used to boost brainpower are detrimental to IQ scores. Using a weak electric current in an attempt to boost brainpower or treat conditions has become popular among scientists and do-it-yourselfers, but a new study shows that using the most common form of electric brain stimulation has a statistically significant detrimental effect on IQ scores.

Strong electric currents may cause a localised lesion in the nervous tissue instead of a functional reversible stimulation. This property has been used for neurosurgical procedures in a variety of treatments such as for Parkinson's disease, focal epilepsy and psychosurgery. Sometimes, the same electrode is used to probe the brain for finding defective functions, before passing the lessening current (electro-coagulation).

A comprehensive review of the research on electrical stimulation of the brain gives a list of different acute impacts of stimulation depending on the brain region targeted. Following are some examples of the effects documented:

Sensory. Feelings of body tingling, swaying, movement, suffocation, burning, shock, warmth, paresthesia, feeling of falling, oscillopsia, dysesthesia, levitation, sounds, phosphenes, hallucinations, micropsia, diplopia and more

Motor. Eye movements, locomotion, speech arrest, automatism,